

The use of calculator and smartphone is prohibited

Exercise 01 (3 points)

For each of the following statements, determine whether the statement is true or false and justify your answer.

1. If P , Q , and R are logical statements such that $((P \vee R) \iff (Q \vee R))$, then $P \iff Q$.
2. For any real numbers x and y , if $xy \neq 1$ and $x \neq y$, then $\frac{x}{x^2 + x + 1} \neq \frac{y}{y^2 + y + 1}$.
3. For any integer $n \geq 2$, we have $\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n-1} + \sqrt{n}} = \sqrt{n} - 1$.
4. If A_1 and A_2 are the sets defined by $A_1 := \left\{1 - \frac{1}{nm} \mid (m, n) \in \mathbb{N} \times \mathbb{N}\right\}$ and $A_2 := \left\{1 - \frac{1}{n(n+1)} \mid n \in \mathbb{N}\right\}$, then $A_1 = A_2$.
5. If A_3 and A_4 are the sets defined by $A_3 := \{(x, y) \in \mathbb{R}^2 \mid 2x + y = 3\}$ and $A_4 := \{(t - 1, 5 - 2t) \mid t \in \mathbb{R}\}$, then $A_3 = A_4$.
6. For any subsets A , B , and C of a set S , we have

$$A \times (B \setminus C) = (A \times B) \setminus (A \times C).$$

Solution :

1. False. If P is any false statement, Q is any true statement, and R is any true statement, then $(P \vee R) \iff (Q \vee R)$ is a true statement, but $P \iff Q$ is a false statement.
2. True. The contrapositive of this statement is

$$\forall x \in \mathbb{R}, \forall y \in \mathbb{R}, \frac{x}{x^2 + x + 1} = \frac{y}{y^2 + y + 1} \implies (xy = 1 \vee x = y).$$

Let x, y be real numbers such that $\frac{x}{x^2 + x + 1} = \frac{y}{y^2 + y + 1}$. This implies $xy^2 + x - x^2y - y = 0$, which leads to $(xy - 1)(y - x) = 0$. Thus, we obtain $xy = 1$ or $x = y$.

3. True. We will prove the property by induction. Since $\frac{1}{\sqrt{1} + \sqrt{2}} = \sqrt{2} - 1$, the property holds for $n = 2$. Assume that the property holds for $n = k$, i.e.,

$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{k-1} + \sqrt{k}} = \sqrt{k} - 1.$$

We have to show that the property holds also for $n = k + 1$. We have

$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{k} + \sqrt{k+1}} = \sqrt{k} - 1 + \frac{1}{\sqrt{k} + \sqrt{k+1}} = \sqrt{k} - 1 + \sqrt{k+1} - \sqrt{k} = \sqrt{k+1} - 1.$$

Thus, the property holds for $n = k + 1$. Therefore, by the principle of mathematical induction, for any integer $n \geq 2$, we have

$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \cdots + \frac{1}{\sqrt{n-1} + \sqrt{n}} = \sqrt{n} - 1.$$

4. False. We have $0 \in A_1$ but $0 \notin A_2$.
5. True. Let $(x, y) \in A_3$. Then, $2x + y = 3$, which implies $y = 3 - 2x$. Set $x = t - 1$, where $t \in \mathbb{R}$. Substituting, we find $y = 5 - 2t$, so $(x, y) = (t - 1, 5 - 2t)$, where $t \in \mathbb{R}$. Therefore, $(x, y) \in A_4$, and we conclude that $A_3 \subseteq A_4$. Now, let $(x, y) \in A_4$. Then $(x, y) = (t - 1, 5 - 2t)$ for some $t \in \mathbb{R}$. The fact that $2x + y = 2(t - 1) + (5 - 2t) = 3$, show that $(x, y) \in A_3$. Thus, $A_4 \subseteq A_3$. Since $A_3 \subseteq A_4$ and $A_4 \subseteq A_3$, we conclude that $A_3 = A_4$.
6. True. Let $(x, y) \in A \times (B \setminus C)$. Then $x \in A$ and $y \in B \setminus C$. This implies $x \in A$, $y \in B$, and $y \notin C$. Since $x \in A$ and $y \in B$, it follows that $(x, y) \in A \times B$. Furthermore, $y \notin C$ implies that $(x, y) \notin A \times C$. Therefore, $(x, y) \in (A \times B) \setminus (A \times C)$. This shows that $A \times (B \setminus C) \subseteq (A \times B) \setminus (A \times C)$.
Now, let $(x, y) \in (A \times B) \setminus (A \times C)$. Then $(x, y) \in A \times B$ and $(x, y) \notin A \times C$. From $(x, y) \in A \times B$, we deduce that $x \in A$ and $y \in B$. From $(x, y) \notin A \times C$ and the fact that $x \in A$, it follows that $y \notin C$. Combining $y \in B$ and $y \notin C$, we conclude that $y \in B \setminus C$. Thus, $x \in A$ and $y \in B \setminus C$ imply $(x, y) \in A \times (B \setminus C)$. This shows that $(A \times B) \setminus (A \times C) \subseteq A \times (B \setminus C)$.

Since we have shown both inclusions, we conclude that $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

Exercise 02 (5 points)

Let f be the map defined by

$$\begin{aligned} f : \mathbb{R} &\longrightarrow \mathbb{R} \\ x &\longmapsto \cos(\pi x). \end{aligned}$$

1. Is f injective? Is it surjective? Justify your answers.
2. Determine the sets $\text{Im } f$, $f(\mathbb{Z})$, $f(\mathbb{R}^+)$, $f^{-1}(\{0\})$, $f^{-1}(\{0, 1\})$, and $f^{-1}(\mathbb{Z})$.

Solution :

1. Since $f(0) = f(2) = 1$ and $0 \neq 2$, it follows that f is not injective. On the other hand, the fact that for any $x \in \mathbb{R}$, $-1 \leq \cos(\pi x) \leq 1$, implies that the number 2 has no preimage by f .
2. Using the fact that $\{\cos t \mid t \in \mathbb{R}\} = [-1, 1]$, we find

$$\text{Im } f = f(\mathbb{R}) = \{\cos(\pi x) \mid x \in \mathbb{R}\} = \{\cos t \mid t \in \mathbb{R}\} = [-1, 1].$$

Similarly, using the fact that $\{\cos t \mid t \in \mathbb{R}\} = \{\cos t \mid 0 \leq t < 2\pi\}$, we have

$$f(\mathbb{R}^+) = \{\cos(\pi x) \mid x \in \mathbb{R}^+\} = \{\cos t \mid t \in [0, 2\pi)\} = [-1, 1].$$

Next, using the fact that for any $x \in \mathbb{Z}$, $\cos(\pi x) \in \{-1, 1\}$, we find

$$f(\mathbb{Z}) = \{\cos(\pi x) \mid x \in \mathbb{Z}\} = \{-1, 1\}.$$

Let θ be a real number. Using the fact that $\cos \theta = 0$ if and only if $\theta = \frac{\pi}{2} + k\pi$ for some $k \in \mathbb{Z}$, $\cos \theta = 1$ if and only if $\theta = 2k\pi$ for some $k \in \mathbb{Z}$, and $\cos \theta = -1$ if and only if $\theta = (2k+1)\pi$ for some $k \in \mathbb{Z}$, we compute the following :

$$f^{-1}(\{0\}) = \{x \in \mathbb{R} \mid \cos(\pi x) = 0\} = \{x \in \mathbb{R} \mid \pi x = \frac{\pi}{2} + k\pi \text{ for some } k \in \mathbb{Z}\} = \left\{ \frac{2k+1}{2} \mid k \in \mathbb{Z} \right\},$$

$$f^{-1}(\{1\}) = \{x \in \mathbb{R} \mid \cos(\pi x) = 1\} = \{x \in \mathbb{R} \mid \pi x = 2k\pi \text{ for some } k \in \mathbb{Z}\} = \{2k \mid k \in \mathbb{Z}\},$$

and

$$f^{-1}(\{-1\}) = \{x \in \mathbb{R} \mid \cos(\pi x) = -1\} = \{x \in \mathbb{R} \mid \pi x = (2k+1)\pi \text{ for some } k \in \mathbb{Z}\} = \{2k+1 \mid k \in \mathbb{Z}\}.$$

From these, we deduce that

$$f^{-1}(\{0, 1\}) = f^{-1}(\{0\}) \cup f^{-1}(\{1\}) = \left\{ \frac{2k+1}{2}, 2k \mid k \in \mathbb{Z} \right\},$$

and

$$\begin{aligned} f^{-1}(\mathbb{Z}) &= \{x \in \mathbb{R} \mid \cos(\pi x) \in \mathbb{Z}\} \\ &= \{x \in \mathbb{R} \mid \cos(\pi x) \in \{-1, 0, 1\}\} \\ &= f^{-1}(\{-1, 0, 1\}) \\ &= f^{-1}(\{-1\}) \cup f^{-1}(\{0\}) \cup f^{-1}(\{1\}) \\ &= \mathbb{Z} \cup \left\{ \frac{2k+1}{2} \mid k \in \mathbb{Z} \right\}. \end{aligned}$$

Exercise 03 (7 points)

1. Let α_1, α_2 , and α_3 be three real numbers. Put $\sigma_1 := \sum_{i=1}^3 \alpha_i$, $\sigma_2 := \sum_{1 \leq i < j \leq 3} \alpha_i \alpha_j$, and $\sigma_3 := \prod_{i=1}^3 \alpha_i$. Show that α_1, α_2 , and α_3 are the solution of the equation $X^3 - \sigma_1 X^2 + \sigma_2 X - \sigma_3 = 0$.
2. Let g be the map defined by

$$\begin{aligned} g: \quad \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \\ (x, y, z) &\longmapsto (x + y + z, xy + xz + yz, xyz). \end{aligned}$$

- (a) Is g injective? Is it surjective? Justify your answers.
 - (b) Let a be a real number. Determine $g^{-1}(\{(a, 0, 0)\})$, $g^{-1}(\{(0, a, 0)\})$, and $g^{-1}(\{(0, 0, a)\})$.
3. Let $h = g|_{\mathbb{Z}^3}$.
 - (a) Determine the set $h^{-1}(\{(2, 1, 0), (0, 0, -2)\})$.
 - (b) By relying on the previous result, examine the injectivity and surjectivity of h .
 4. Let $U := \{(x, y, z) \in \mathbb{R}^3 \mid x < y < z \text{ and } x + y + z = 0\}$. Determine explicitly $V \subset \mathbb{R}^3$ so that the map

$$\begin{aligned} \tilde{g}: \quad U &\longrightarrow V \\ (x, y, z) &\longmapsto g(x, y, z) \end{aligned}$$

is bijective.

Solution :

1. Since $(X - \alpha_1)(X - \alpha_2)(X - \alpha_3) = X^3 - \sigma_1 X^2 + \sigma_2 X - \sigma_3$, then α_1, α_2 and α_3 are the solution of the equation $X^3 - \sigma_1 X^2 + \sigma_2 X - \sigma_3 = 0$.

2. (a) Since $g(1, 0, 0) = (1, 0, 0) = g(0, 1, 0)$, then g is not injective. On the other hand, assume that there exist real numbers x, y, z such that $g(x, y, z) = (0, 1, 0)$, then x, y and z are the solutions of the equation $X^3 + X = 0$. But the solutions of this equation are $0, i$, and $-i$. Thus, for any $(x, y, z) \in \mathbb{R}^3$, $g(x, y, z) \neq (0, 1, 0)$, implying that g is not surjective.

(b) Let x, y , and z be real numbers such that $g(x, y, z) = (a, 0, 0)$. Then, x, y , and z are the solutions of the equation $X^2(X - a) = X^3 - aX^2 = 0$. Thus, we deduce that

$$g^{-1}(\{(a, 0, 0)\}) = \{(a, 0, 0), (0, a, 0), (0, 0, a)\}.$$

Next, let x', y' , and z' be real numbers such that $g(x', y', z') = (0, a, 0)$. Then, x', y' , and z' are the solutions of the equation $X(X^2 + a) = X^3 + aX = 0$. From this, we deduce that

$$g^{-1}(\{(0, a, 0)\}) = \begin{cases} \emptyset & \text{if } a > 0, \\ \{(0, 0, 0)\} & \text{if } a = 0, \\ \{(0, \sqrt{-a}, -\sqrt{-a}), (0, -\sqrt{-a}, \sqrt{-a}), (\sqrt{-a}, 0, -\sqrt{-a}), \\ \quad (-\sqrt{-a}, 0, \sqrt{-a}), (\sqrt{-a}, -\sqrt{-a}, 0), (-\sqrt{-a}, \sqrt{-a}, 0)\} & \text{if } a < 0. \end{cases}$$

Finally, let x'', y'' , and z'' be real numbers such that $g(x'', y'', z'') = (0, 0, a)$. Then, x'', y'' , and z'' are the solutions of the equation $X^3 - a = 0$. Since this equation admits only one real solution, we deduce that

$$g^{-1}(\{(0, 0, a)\}) = \emptyset.$$

3. First, consider the equation $h(x, y, z) = (2, 1, 0)$. This implies that x, y , and z are the solutions of the equation $X(X - 1)^2 = X^3 - 2X^2 + X = 0$. Hence, $h^{-1}(\{(2, 1, 0)\}) = \{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$. Next, consider the equation $h(x, y, z) = (0, 0, -2)$. This implies that x, y , and z are the solutions of the equation $X^3 + 2 = 0$. Since this equation has no integer solutions, it follows that $h^{-1}(\{(0, 0, -2)\}) = \emptyset$. Therefore,

$$h^{-1}(\{(2, 1, 0), (0, 0, -2)\}) = \{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}.$$

(b) The fact that $\#h^{-1}(\{(2, 1, 0)\}) = 3$ implies that h is not injective, while $\#h^{-1}(\{(0, 0, -2)\}) = 0$ shows that h is not surjective.

4. The constraint $x < y < z$ imposed on the elements of U ensures that $\tilde{g}$ is injective. Consequently, $\tilde{g}$ is bijective if and only if it is surjective.

$\tilde{g}$ is surjective if and only if for every $(\ell, p, q) \in V$, the equation $(\ell, p, q) = \tilde{g}(x, y, z)$ admits at least one solution $(x, y, z) \in U$. This is equivalent to requiring that $\ell = 0$ and the cubic equation $X^3 + pX - q = 0$ has

three distinct real solutions. The latter condition holds if and only if $3X^2 + p = 0$ has two distinct real solutions x_1 and x_2 , satisfying $(x_1^3 + px_1 - q)(x_2^3 + px_2 - q) < 0$. This is equivalent to requiring $p < 0$, and $\left(\left(\sqrt{\frac{-p}{3}}\right)^3 + p\sqrt{\frac{-p}{3}} - q\right)\left(\left(-\sqrt{\frac{-p}{3}}\right)^3 - p\sqrt{\frac{-p}{3}} - q\right) < 0$, which means $4p^3 + 27q^2 < 0$. Thus, the set V is given by :

$$V = \{(0, p, q) \in \mathbb{R}^3 \mid 4p^3 + 27q^2 < 0\}.$$